

An Analysis of Autoregressive model

Autoregressive model

Autoregressive model -- Part 1

$$X_t = \phi X_{t-N} + \sum_{k=0}^{N-1} \phi_k \varepsilon_{t-k} \quad \{\displaystyle X_t = \varphi^N X_{t-N} + \sum_{k=0}^{N-1} \varphi_k \varepsilon_{t-k}\}$$

Autoregressive model -- Part 2

Here predicted values of X_t would be based on the p future values of the same series. This way of estimating the AR parameters is due to John Parker Burg, and is called the Burg method: Burg and later authors called these particular estimates "maximum entropy estimates", but the reasoning behind this applies to the use of any set of estimated AR parameters. Compared to the estimation scheme using only the forward prediction equations, different estimates of the autocovariances are produced, and the estimates have different stability properties. Burg estimates are particularly associated with maximum entropy spectral estimation. Other possible approaches to estimation include maximum likelihood estimation. Two distinct variants of maximum likelihood are available: in one (broadly equivalent to the forward prediction least squares scheme) the likelihood function considered is that corresponding to the conditional distribution of later values in the series given the initial p values in the series; in the second, the likelihood function considered is that corresponding to the unconditional joint distribution of all the values in the observed series. Substantial differences in the results of these approaches can occur if the observed series is short, or if the process is close to non-stationarity.

Autoregressive model -- Part 3

An autoregressive model can thus be viewed as the output of an all-pole infinite impulse response filter whose input is white noise. Some parameter constraints are necessary for the model to remain weak-sense stationary. For example, processes in the AR(1) model with $|\phi| \geq 1$ $\{\displaystyle |\varphi| \geq 1\}$ are not stationary. More generally, for an AR(p) model to be weak-sense stationary, the roots of the polynomial $\Phi(z) := 1 - \sum_{i=1}^p \phi_i z^i$ $\{\displaystyle \Phi(z) := 1 - \sum_{i=1}^p \varphi_i z^i\}$ must lie outside the unit circle, i.e., each (complex) root z_i $\{\displaystyle z_i\}$ must satisfy $|z_i| > 1$ $\{\displaystyle |z_i| > 1\}$ (see pages 89,92).

Autoregressive model -- Part 4

where B is the backshift operator, where $\phi(\cdot)$ $\{\displaystyle \varphi(\cdot)\}$ is the function defining the autoregression, and where ϕ_k $\{\displaystyle \varphi_k\}$ are the coefficients in the autoregression. The formula is valid only if all the roots have

multiplicity 1. The autocorrelation function of an AR(p) process is a sum of decaying exponentials.

Autoregressive model -- Part 5

have been estimated, the autoregression can be used to forecast an arbitrary number of periods into the future. First use t to refer to the first period for which data is not yet available; substitute the known preceding values X_{t-i} for $i=1, \dots, p$ into the autoregressive equation while setting the error term ε_t $\{\displaystyle \varepsilon_t\}$ equal to zero (because we forecast X_t to equal its expected value, and the expected value of the unobserved error term is zero). The output of the autoregressive equation is the forecast for the first unobserved period. Next, use t to refer to the next period for which data is not yet available; again the autoregressive equation is used to make the forecast, with one difference: the value of X one period prior to the one now being forecast is not known, so its expected value—the predicted value arising from the previous forecasting step—is used instead. Then for future periods the same procedure is used, each time using one more forecast value on the right side of the predictive equation until, after p predictions, all p right-side values are predicted values from preceding steps. There are four sources of uncertainty regarding predictions obtained in this manner: (1) uncertainty as to whether the autoregressive model is the correct model; (2) uncertainty about the accuracy of the forecasted values that are used as lagged values in the right side of the autoregressive equation; (3) uncertainty about the true values of the autoregressive coefficients; and (4) uncertainty about the value of the error term ε_t $\{\displaystyle \varepsilon_t\}$ for the period being predicted. Each of the last three can be quantified and combined to give a confidence interval for the n -step-ahead predictions; the confidence interval will become wider as n increases because of the use of an increasing number of estimated values for the right-side variables.

Autoregressive model -- Part 6

The simplest AR process is AR(0), which has no dependence between the terms. Only the error/innovation/noise term contributes to the output of the process, so in the figure, AR(0) corresponds to white noise. For an AR(1) process with a positive ϕ $\{\displaystyle \varphi\}$, only the previous term in the process and the noise term contribute to the output. If ϕ $\{\displaystyle \varphi\}$ is close to 0, then the process still looks like white noise, but as ϕ $\{\displaystyle \varphi\}$ approaches 1, the output gets a larger contribution from the previous term relative to the noise. This results in a "smoothing" or integration of the output, similar to a low pass filter. For an AR(2) process, the previous

An Analysis of Autoregressive model

Autoregressive model

two terms and the noise term contribute to the output. If both ϕ_1 and ϕ_2 are positive, the output will resemble a low pass filter, with the high frequency part of the noise decreased. If ϕ_1 is positive while ϕ_2 is negative, then the process favors changes in sign between terms of the process. The output oscillates. This can be linked to edge detection or detection of change in direction.

Autoregressive model -- Part 7

If $\phi_1 > 0$ there is a single spectral peak at $f = 0$, often referred to as red noise. As ϕ_1 becomes nearer 1, there is stronger power at low frequencies, i.e. larger time lags. This is then a low-pass filter, when applied to full spectrum light, everything except for the red light will be filtered. If $\phi_1 < 0$ there is a minimum at $f = 0$, often referred to as blue noise. This similarly acts as a high-pass filter, everything except for blue light will be filtered.